Zhengyan Lyu

+ 1 401 215 3842 | zhengyan_lyu@brown.edu | https://imbalzy.github.io/

EDUCATION:

Brown University, *Providence, RI* Expected: December 2021

- Master of Science in Computer Science
- Coursework: Database Systems, Computer Linguistics, Deep Learning, Distributed System, Computation in Economics and Games
- Overall GPA: **3.8**

Villanova University, *Villanova, PA* August 2017-May 2020

- Bachelor of Science in Computer Science Minor: Mathematics
- Coursework: Database Systems, Advanced Algorithms, Deep Learning, Linear Algebra, Computer System, Programming Languages, Theory of Computability, Software Engineering
- Major GPA:**3.84**

SKILLS:

- **Programming languages:** Familiar with **Python** (4 years+), **Java** (4 years+), **JavaScript** (2 years+)**.** Experience with C/C++, Go, SQL, Html, CSS, Lisp, TypeScript
- **Technical skills**: **Vue**, **Pytorch, TensorFlow**, **Linux**, React.js, Git, Node.js, Scikit-learn, SVN

WORK EXPERIENCE:

Full-stack Software Engineering Intern, *TuSimple, Remote* May 2021-August 2021

- Developed self-driving car simulation systems for both front end and back end
- Constructed simulation management webpage with **HTML/CSS** and **JavaScript** with **Vue 3.0** framework at front end
- Processed/enveloped data with **Python** and **tsRPC** and queried with **PostgreSQL+PonyOrm** at back end

Full-stack Software Engineering Intern, *Shengqu Games, Shanghai* June 2020-November 2020

- Developed both front end and back end of game master systems for two mobile games independently
- **HTML/CSS** and **JavaScript** with **Vue.js 2.0** at front end, **Node.js** at back end, **MongoDB** for database management system
- Refactored part of the code base with **ViewUI** toolkit and packaged **Vue** components to replace old duplicated code for better code readability and modifiability

PROJECTS/ RESEARCH:

Latent Space Arithmetic in Deep Learning, *Brown University, Providence, RI* October 2020-January 2021

- Constructed a GAN model to generate images and modify attributes of images in **Python** with **Tensorflow**
- Proposed a new loss function to solve the problem of disentanglement in latent space arithmetic

Deep learning Research Assistant, *Villanova University, Villanova, PA* May 2019-August 2019

- Constructed a weapon detection model in **Python** with **Pytorch** library
- Implemented the model on preprocessed weapon datasets by transfer learning and fine-tuning
- Presented the poster with the **95%** accuracy model at Villanova University in September 2019

E-Waste Trading System, *Villanova University*, *Villanova, PA* September 2018-December 2018

- Designed login and search user interfaces in **React, HTML/CSS** and **JavaScript**
- Stored data with **MySQL** relational database system using phpMyAdmin administration tool
- Edited all code on **Linux** Raspberry Pi remotely and ran **Apache** HTTP Server

CONTESTS AND ACTIVITIES:

Programming Contest Team, *Villanova, PA* September 2017-May 2020

- **15th** place in **ICPC**(International Collegiate Programming Contest) Mid-Atlantic Region 2019
- Hosted internal programming contest and presented solutions to team members in meeting every week